

Products – Control (PID Tuning)

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Configures the PID controller that adjusts the motor or valve to maintain the target rate.

Primary Settings

Setting	Description
1. Gain	Rate adjustment factor — how strongly the controller responds to rate errors. The effect is linear: doubling Gain doubles the correction. Errors are measured as a percentage of the target rate, so the same Gain setting behaves consistently at low and high rates.
2. Integral	Trims residual error once the rate is close to target (inside the BrakePoint band — see below). Far from target, only Gain (proportional) drives the approach; Integral does not speed that up. Increase if the rate settles near, but not exactly on, target. Integral accumulates once per PID loop tick, so a shorter PID loop time (setting 4 below) also makes it trim faster in real time, independent of this setting.
3. Max Power	Maximum PWM power delivered to motor or valve
4. Min Power	Minimum PWM power delivered to motor or valve

Adjustment Process

1. **Initial Setup** — Set minimum power to the minimum needed to move the motor/valve. Set maximum power to 100. Set Integral to 0.
2. **Gain Adjustment** — Slowly increase gain until the system overshoots the target, then reduce slightly to stabilize.
3. **Integral** — If the rate approaches target but settles slightly off, increase Integral to close the remaining gap. It only acts once you're already close (see BrakePoint) — it will not speed up a slow ramp toward target.

Optional / Advanced Settings

Settings 7–9 apply to the Combo Timed control type only; setting 5 applies to Motor only.

Setting	Description
1. Deadband	Error % below which no adjustment is made. Also controls when accumulated Integral is reset — only once the error clearly crosses to the other side of target, not on small wobble near the setpoint.
2. BrakePoint	Error % (of peak target) defining the "near target" band. Outside this band, only fast proportional (Gain) control acts and Integral stays at zero. Inside it, proportional gain drops to the Slow Adjustment % and Integral begins accumulating.
3. Slow Adjustment	% of full proportional adjustment used once inside the BrakePoint band
4. PID loop time	How often (50–250 ms) the PID recalculates its output. A shorter time makes the valve/motor react sooner to a real rate change, and also speeds up Integral trimming in real time (see Integral, above) since it accumulates once per tick. Reducing this after Gain/Integral were tuned at a slower rate can introduce new oscillation — check for overshoot as with any Gain increase.
5. Slew Rate	Motor control type only — maximum total PWM change per PID loop. Has no effect on Standard/Combo Close valves, which are positioned fresh each loop rather than ramped.
6. Max Integral change	Maximum Integral PWM change per PID loop. The total accumulated Integral is separately capped to the Max Power – Min Power range, so it can reach full authority (e.g. to free a stuck valve) without jumping there in one loop.
7. Min Start %	Combo Timed control type only — % of target rate below which the adjust/pause cycle is skipped for a faster start from 0.
8. Adjust time	Combo Timed control type only — milliseconds the valve is actively adjusted during each cycle.
9. Pause time	Combo Timed control type only — milliseconds the valve pauses (holds) between adjustments.
10. Min flow sensor Hz	Pulse frequency below which the sensor is treated as stopped/no-flow
11. Max flow sensor Hz	Pulse frequency above which pulses are rejected as sensor noise/bounce
12. Flow sample time (Smp Tm)	Time window in milliseconds (50–1500) used to calculate the median flow reading. A longer window smooths the reading; a shorter window responds faster. Default 400 ms.

Controls

Control	Action
 Navigate left	Previous product
 Navigate right	Next product
 Load defaults	Restore default PID values
 Open rate graph	Display the rate graph
 Gain +	Increase Gain value
 Gain -	Decrease Gain value
 Integral +	Increase Integral value
 Integral -	Decrease Integral value
 Max Power +	Increase Max Power value
 Max Power -	Decrease Max Power value
 Min Power +	Increase Min Power value
 Min Power -	Decrease Min Power value
 Show advanced settings	Expand the advanced/optional settings panel
 Cancel	Discard changes
 Save	Save settings